

Lesson Overview: Introduction to ETL

Understanding data pipelines is essential for data engineers to effectively manage the flow and transformation of data. Mastering these concepts is foundational to supporting data analytics and driving insights within any organization. This lesson series equips us with the necessary skills to build, optimize, and maintain data pipelines, ensuring data is accurate, secure, and readily available for analysis. We'll explore the fundamentals of data pipelines, including their architecture, components, and workflows. We'll take a look the ETL process, how to handle different data types, and understand the costs involved in building pipeline and delve into the principles of constructing great data pipelines, offering insights into best practices and tool selection.



What You'll Learn:

Data Pipeline Basics

Understand the fundamental workings of data pipelines and their roles in data management.

Pipeline Architecture & ETL

Understand the architecture and operation of data pipelines and the ETL process.

Pipeline Components & Stages

Identify and work with different components and stages of data pipelines.

Data Collection & Ingestion

Learn techniques for collecting and ingesting data efficiently into pipeline systems.

Data Sharing & Consumption

Explore strategies for sharing and consuming data across the business and with external stakeholders.

Data Flow & Transformation Management

Manage data flow and transformation with best practices.

XML Processing

Process XML data and convert it into structured formats.

Cost Considerations

Assess the costs of pipeline construction and operation.

Pipeline Tools & Techniques

Discover tools and techniques for efficient data pipeline implementation.